



Taha
Zeeshan
Software Engineer , B.Sc.

[in](#) [Linkedin Profile](#)

+49-152-15458567

tahazeeshan09@gmail.com

Max-Planck-Straße, Heilbronn

Date-of-birth: 12.01.2005

About Me

Software Engineer specializing in Autonomous Systems and Computer Vision. My experience ranges from developing ROS2 navigation pipelines for Mars Rovers to deploying YOLOv8 models for real-time agricultural monitoring. I'm most at home at the intersection of hardware and software—optimizing CNNs for edge devices or building sensor-fusion logic for AI wearables.

Projects

1 September 2023-
1 January 2026

Team Leader | Autonomous system Engineer at Ulurover , Bursa , Turkey

- Engineered a vision-based localization pipeline by implementing ORB-SLAM3 and ZED2 stereo vision for real-time spatial feature extraction for **Mars Rover Team**.
- Developed sensor fusion algorithms utilizing Extended Kalman Filters (EKF) to integrate high-frequency IMU data with visual odometry for precise state estimation.
- Directed a multidisciplinary team of 25+ engineers while securing 250k TL in corporate sponsorships, establishing the project as the highest-funded technical team at the university.

Software Head | Computer Vision Engineer at Itech , Bursa , Turkey

- Architected a real-time computer vision system for AI wearables, utilizing Deep Learning models for multi-class obstacle detection and scene recognition.
- Integrated Natural Language Processing (NLP) for voice-controlled system interaction, ensuring low-latency response times for visually impaired users.
- Trained custom vision models for edge deployment, securing 3rd place at the Youth Tech Innovators 2025 competition for technical innovation.

Software Head | Computer Vision Engineer at TarimTek , Bursa , Turkey

- Trained and deployed a YOLOv8 object detection model using Convolutional Neural Networks (CNN) for real-time insect classification in outdoor environments.
- Curated a custom annotated dataset using OpenCV, optimizing the vision pipeline for variable lighting and high-noise field conditions.
- Refined model inference speeds to ensure robust deployment on edge devices, maintaining high detection accuracy during active field operations.

Education

Bachelors in Software
Engineering
March 2026-present

Bachelors in Computer
Engineering
October 2024 - January 2026

Hochschule Heilbronn , Germany (Exchange Program)
B.S. in Software Engineering

Bursa uludag University, Turkey
B.S. in Computer Engineering

Certifications

2025

CS50 Python Programming , Harvard University

David J. Milan

2025

Advanced Machine learning Algorithms , Stanford

Andrew Ng

2024

Supervised Machine Learning , Stanford

Andrew Ng

2024

Python Basics , University of Michigan

Paul Resnick

SCHOLARSHIPS & AWARDS

Erasmus plus Scholarship

Awarded erasmus plus scholarship for doing exchange semester of Bachelors in Software Engineering studies at Hochschule Heilbronn, Germany.

Turkey Burslari Scholarship

Awarded a fully-funded scholarship for doing Bachelors in Computer Engineering studies at Bursa Uludağ University, Turkey.

Youth Tech Begin Innovators 2025:

Secured third place and a 30,000 TL prize for the iGlasses smart assistive technology project in Youth Tech Begin 2025 Competition.

Pakistan National Board Exams (BISE):

Secured 100% marks and ranked as the National Top Performer in board examinations.

Skills & Interests

Programming Languages: Python (Advanced), C++, C#, JavaScript, ROS2 , ROS

Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn.

Computer Vision & Perception: YOLOv8, OpenCV (Image Processing), CNNs, OCR, Stereo Vision (ZED2), EfficientNet.

Sensors: RplidarA1 , Zed-mini , Zed-2 , Raspberry-pi, Jetson Orin Nano, imx219-83 stereo camera, Pixhawk

Languages

- English - C2
- Turkish - C2
- German - A1
- Urdu - Native
- Hindi - Native

Interests

- Providing technical support and mobility assistance for local accessibility projects helping people with disabilities.